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PHYSICS

9203/2

Paper 2

Mark scheme

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2 5 6 Y 9 2 0 3 / 2 / M S

Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Examiner.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

Further copies of this mark scheme are available from oxfordaqa.com

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Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 2 with a small amount of level 3 material it would be placed in level 2 but be awarded a mark near the top of the level because of the level 3 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Information to Examiners

1. General

The mark scheme for each question shows:
- the marks available for each part of the question
- the total marks available for the question
- the typical answer or answers which are expected
- extra information to help the Examiner make his or her judgement
- the Assessment Objectives, level of demand and specification content that each question is intended to cover.

The extra information is aligned to the appropriate answer in the left-hand part of the mark scheme and should only be applied to that item in the mark scheme.

At the beginning of a part of a question a reminder may be given, for example: where consequential marking needs to be considered in a calculation; or the answer may be on the diagram or at a different place on the script.

In general the right-hand side of the mark scheme is there to provide those extra details which confuse the main part of the mark scheme yet may be helpful in ensuring that marking is straightforward and consistent.

2. Emboldening and underlining

2.1 In a list of acceptable answers where more than one mark is available 'any **two** from' is used, with the number of marks emboldened. Each of the following bullet points is a potential mark.

2.2 A bold **and** is used to indicate that both parts of the answer are required to award the mark.

2.3 Alternative answers acceptable for a mark are indicated by the use of **or**. Different terms in the mark scheme are shown by a / ; eg allow smooth / free movement.

2.4 Any wording that is underlined is essential for the marking point to be awarded.

3. Marking points

3.1 Marking of lists

This applies to questions requiring a set number of responses, but for which students have provided extra responses. The general principle to be followed in such a situation is that 'right + wrong = wrong'.

Each error / contradiction negates each correct response. So, if the number of errors / contradictions equals or exceeds the number of marks available for the question, no marks can be awarded.

However, responses considered to be neutral (indicated as * in example 1) are not penalised.

Example 1: What is the pH of an acidic solution?

[1 mark]

Student	Response	Marks awarded
1	green, 5	0
2	red*, 5	1
3	red*, 8	0

Example 2: Name two planets in the solar system.

[2 marks]

Student	Response	Marks awarded
1	Neptune, Mars, Moon	1
2	Neptune, Sun, Mars, Moon	0

3.2 Use of chemical symbols/formulae

If a student writes a chemical symbol/formula instead of a required chemical name, full credit can be given if the symbol/formula is correct and if, in the context of the question, such action is appropriate.

3.3 Marking procedure for calculations

Marks should be awarded for each stage of the calculation completed correctly, as students are instructed to show their working. Full marks can, however, be given for a correct numerical answer, without any working shown.

3.4 Interpretation of ‘it’

Answers using the word ‘it’ should be given credit only if it is clear that the ‘it’ refers to the correct subject.

3.5 Errors carried forward

Any error in the answers to a structured question should be penalised once only.

Papers should be constructed in such a way that the number of times errors can be carried forward is kept to a minimum. Allowances for errors carried forward are most likely to be restricted to calculation questions and should be shown by the abbreviation ecf in the marking scheme.

3.6 Phonetic spelling

The phonetic spelling of correct scientific terminology should be credited **unless** there is a possible confusion with another technical term.

3.7 Brackets

(…..) are used to indicate information which is not essential for the mark to be awarded but is included to help the examiner identify the sense of the answer required.

3.8 Allow

In the mark scheme additional information, ‘allow’ is used to indicate creditworthy alternative answers.

3.9 Ignore

Ignore is used when the information given is irrelevant to the question or not enough to gain the marking point. Any further correct amplification could gain the marking point.

3.10 Do <u>not</u> accept

Do **not** accept means that this is a wrong answer which, even if the correct answer is given as well, will still mean that the mark is not awarded.

Question 1

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.1	the driver being distracted the tiredness of the driver		1 1	AO1 3.1.5c 1–3

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.2	the speed of the car		1	AO1 3.1.5b 4–5

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.3	friction		1	AO1 3.1.5e 1–3

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.4	$F = 1200 \times 3.0$ $F = 3600$ (N)		1 1	AO2 3.1.3h 1–3

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.5	allow any **one** from: • brakes overheating • loss of control	 allow the car may skid	1	AO1 3.1.5b 1–3

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.6	the studs increase friction (between tyres and snow)	allow grip for friction	1	AO3 3.1.5d 1–3

Total Question 1		**8**

Question 2

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.1	$F = 200 \times 0.025$		1	AO2 3.1.1h 1–3
	$F = 5$ (N)	allow 5.0 (N)	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.2	$0.0049 = 0.005 \times 9.8 \times h$		1	AO2 3.2.1d 4–5
	$h = \frac{0.0049}{0.005 \times 9.8}$		1	
	$h = 0.1$ (m)	allow 0.10 (m)	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.3	the further the handle is pulled down the greater the compression		1	AO3 3.1.1fg 3.2.2a 4–5
	so elastic potential energy stored is greater		1	
	therefore the kinetic energy of the ball is greater		1	
	and gravitational potential energy is greater (and the vertical height of the ball is greater)		1	

Total Question 2		**9**

Question 3

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.1	a change in the speed of the light wave		1	AO1 3.3.1f 1–3

Question	Answers	Mark	AO/ Spec. Ref.
03.2	**Level 3:** The method would lead to the production of a valid outcome. All key steps are identified and logically sequenced.	5–6	AO4 3.3.5e 4–5
	Level 2: The method would not necessarily lead to a valid outcome. Most steps are identified, but the method is not fully logically sequenced.	3–4	
	Level 1: The method would not lead to a valid outcome. Some relevant steps are identified, but links are not made clear.	1–2	
	No relevant content	0	
	Indicative content Some indicative content could be indicated within a labelled diagram. • place a glass block on a piece of paper • draw around the glass block • use a protractor draw a line normal to the surface • use a protractor to measure an angle of incidence of 30 degrees from the normal • use the ray box to shine a ray of light through the glass block at 30 degrees • mark the ray of light emerging from the glass block • join the points to show the path of the complete ray through the block • and draw a line normal to the surface • use a protractor to measure the angle of refraction from the normal line • use a ray box to shine a ray of light at a range of different angles (of incidence) • increase the angle of incidence in 10 degree intervals • at angles of incidence from 30 to 80 degrees Methods involving mirrors and reflection score zero		

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.3	$n = \dfrac{\sin 40°}{\sin 23°}$		1	AO2 3.3.5e 4–5
	n = 1.645 …	allow answer to 2 or more significant figures	1	
	n = 1.6		1	

Total Question 3		10

Question 4

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.1	**C**		1	AO1 3.3.1g 1–3

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.2	**D**		1	AO1 3.3.1g 1–3

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.3	minimum and maximum (values)	allow difference between the maximum and minimum values allow lowest to highest (value)	1	AO4 3.3.1g 4–5

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.4	number of waves (passing a point) in one second		1	AO1 3.3.1g 4–5

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.5	black surfaces are better emitters of infrared radiation		1	AO4 3.3.2d 4–5

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.6	the higher the temperature of water, the greater the frequency of the infrared radiation		1	AO1 3.3.2e 4–5

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.7	$3.0 \times 10^8 = 0.00075 \times \lambda$		1	AO2 3.3.1h 6–7
	$f = \dfrac{3.0 \times 10^8}{0.00075}$	allow a correct rearrangement using an incorrectly / not converted value of wavelength	1	
	$f = 4.0 \times 10^{11}$ (Hz)	allow a correct calculation using an incorrectly / not converted value of wavelength	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.8	all objects emit infrared radiation		1	AO3 3.3.2h 6–7

Total Question 4		**10**

Question 5

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.1	gravitational force	allow gravity	1	AO1 3.8.1a 1–3

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.2	fusion		1	AO1 3.8.1b 4–5

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.3	forces are balanced	allow the forces are equal and opposite	1	AO1 3.8.1b 4–5

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.4	10.5 (thousands of millions of years)	allow $\pm \frac{1}{2}$ small square	1	AO2 3.8.1f 1–3

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.5	as the mass of the star increases, the time spent on the main sequence decreases		1	AO3 3.8.1e 4–5
	at a decreasing rate	allow it is non-linear	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.6	a star that becomes a black hole has a greater mass than a star that becomes a black dwarf		1	AO3 3.8.1ej 6–7
	therefore spends less time as a main sequence star	dependent on MP1	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.7	elements heavier than iron are only formed during a supernova		1	AO1 3.8.1i 6–7
	the mass of the Sun is not great enough for a supernova to occur		1	

Total Question 5		**10**

Question 6

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.1	swimmer **B**		1	AO2 3.1.2c 1–3

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.2	$1.6 = \frac{s}{125.00}$ $s = 1.6 \times 125.00$ $s = 200$ (m)		1 1 1	AO2 3.1.2c 4–5

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.3	the smallest change / measurement / time that can be detected	allow smallest measurement that can be recorded by the instrument	1	AO1 3.1.2c 6–7

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.4	both swimmers **A** and **D** would have the same time		1	AO3 3.1.2c 6–7

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.5	there is a delay in starting the stopclock as the sound waves travel from the starting pistol to the person's ear		1	AO4 3.1.2c 6–7
	there is a reaction time from hearing the sound to starting the stopclock		1	
	there is uncertainty at the end of the race	allow there is a reaction time at the end of the race from when the person sees the swimmer finish to stopping the stopclock	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.6	m = 0.080 (kg)		1	AO2 3.1.4c 3.2.1e 8–9
	0.090 = 0.5 × 0.080 × v^2	allow correct substitution using incorrectly / not converted mass	1	
	v = 1.5 (m/s)	allow a correct calculation using an incorrectly / not converted value of mass	1	
		for subsequent marks to be awarded the correct equation must have been used to calculate the velocity		
	$F = \dfrac{1.5 \times 0.080}{2.5}$	allow a correct substitution using an incorrectly calculated value of velocity	1	
	F = 0.048 (N)	allow a correct calculation using an incorrectly calculated value of velocity	1	
	and the water does not activate the timer			

Total Question 6		**14**

Question 7

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.1	**(similarity)** both have 6 protons **(difference)** carbon-14 has 2 more neutrons	allow same number of protons allow carbon-14 has 8 neutrons and carbon 12 has 6 neutrons	1 1	AO2 3.7.1f 1 × 4–5 1 × 6–7

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.2	a neutron decays into a proton a (high energy) electron is emitted from the nucleus		1 1	AO1 3.7.2e 6–7

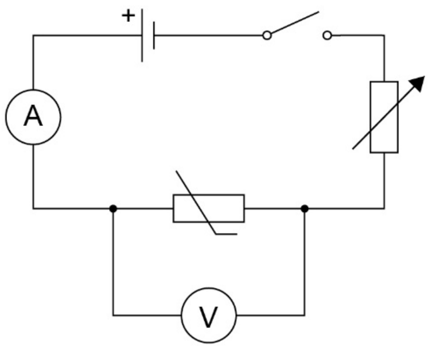
Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.3	${}^{14}_{6}\mathrm{C} \longrightarrow {}^{14}_{7}\mathrm{N} + {}^{0}_{-1}\beta$	**1** mark for the top row **1** mark for the bottom row if no other mark awarded allow **1** mark for the correct numbers for the beta particle	2	AO2 3.7.2f 6–7

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.4	beta radiation is ionising carbon-14 has a very long half-life so will stay radioactive for thousands of years	allow beta radiation can cause cancer	1 1	AO3 3.7.2j 3.7.3e 6–7

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.5	**Level 2:** Scientifically relevant features are identified; the way(s) in which they are similar / different is made clear and (where appropriate) the magnitude of the similarity/difference is noted.		3–4	AO3 3.6.5b 2 × 4-5 2 × 6-7
	Level 1: Relevant features are identified and differences noted.		1–2	
	No relevant content		0	
	Indicative content **advantages of AA battery** • does not produce ionising radiation • does not need shielding to stop beta particles leaving the battery • provides more energy per second so can be used for more applications **disadvantages of AA battery** • lifetime is shorter so needs replacing often • does not provide as much energy so needs replacing more often **advantages of carbon-14 battery** • does not need replacing as battery will last longer than anything it is being used for • can be used for remote applications as the battery does not need replacing • can provide a large amount of energy over its lifetime • allows radioactive waste to be recycled **disadvantages of carbon-14 battery** • needs shielding so beta particles are not emitted • low power rating so limited applications			

Total Question 7		**12**

Question 8

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.1	ammeter drawn in series with thermistor voltmeter drawn in parallel with thermistor thermistor in series with cell and variable resistor	if no other mark awarded allow **1** mark for correct symbols of ammeter, voltmeter and thermistor  full marks can be awarded if a switch is not seen	1 1 1	AO4 3.5.1g 6–7

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.2	to change the (total) resistance of the circuit **or** to change the potential difference across the thermistor		1	AO4 3.5.1h 4–5

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.3	0.028 (mA) 2.8×10^{-2} (mA)	 allow incorrect reading in standard form	1 1	AO2 3.5.1c 6–7

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.4	a large current would cause the thermistor to heat up the resistance of the thermistor would then decrease		1 1	AO3 3.5.1k 6–7

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.5	current is directly proportional to potential difference	allow for **1** mark as the potential difference increases the current increases	2	AO3 3.5.1h 6–7

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.6	current in thermistor = 0.080 mA		1	AO2 3.5.1h 8–9
	current in resistor = (0.13 mA– 0.080 mA) = 0.050 mA	allow a correct calculation using an incorrect value of current in thermistor	1	
	4.3 = 0.000050 × R	allow a correct substitution using an incorrectly calculated value of current	1	
	$R = \frac{4.3}{0.000050}$	allow a correct rearrangement using an incorrectly calculated value of current	1	
	R = 86 000 (Ω)	allow a correct calculation using an incorrectly calculated value of current	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.7	the gradient would not be constant because the resistance would change	allow the graph would not be a straight line dependent on MP1	1 1	AO3 3.5.1k 1 × 6–7 1 × 8–9

Total Question 8		**17**